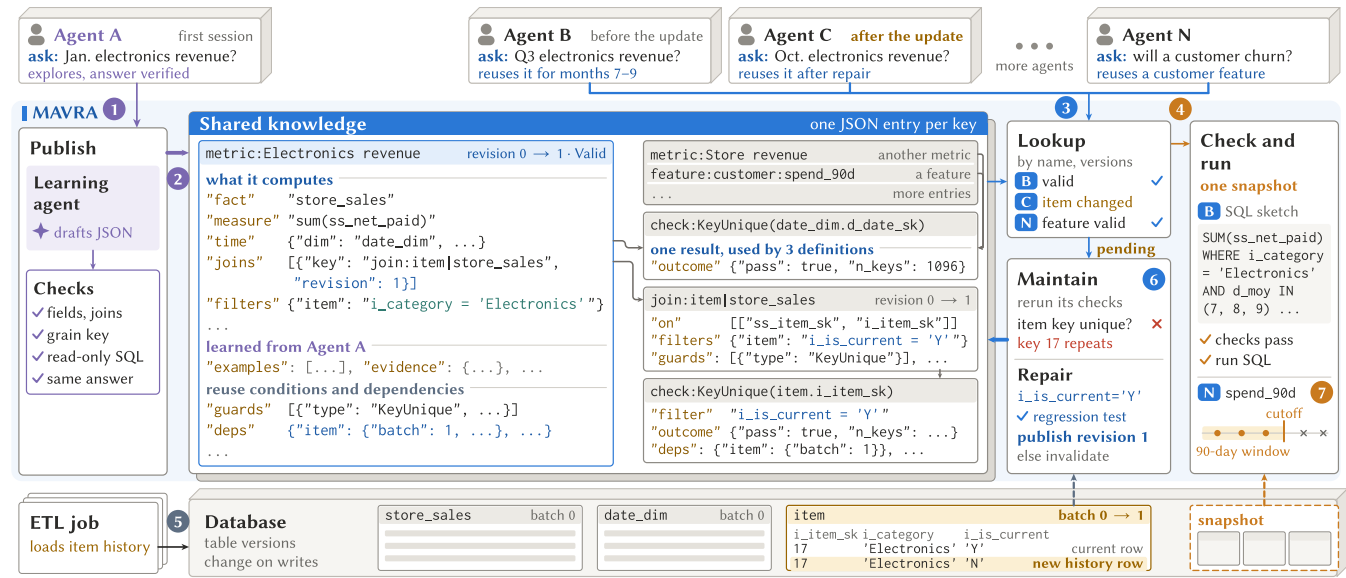

Agent A
first session
ask: Jan. electronics revenue?
explores, answer verified
Agent B
before the update
ask: Q3 electronics revenue?
reuses it for months 7–9
Agent C
after the update
ask: Oct. electronics revenue?
reuses it after repair
more agents
Agent N
ask: will a customer churn?
reuses a customer feature
MAVRA
1
Publish
Learning agent
drafts JSON
Checks
fields, joins
grain key
read-only SQL
same answer
2
Shared knowledge
one JSON entry per key
metric:Electronics revenue
revision 0 → 1 · Valid
what it computes
"fact" "store_sales"
"measure" "sum(ss_net_paid)"
"time" {"dim": "date_dim", ...}
"joins" [{"key": "join:item|store_sales",
"revision": 1}]
"filters" {"item": "i_category = 'Electronics'"}
...
learned from Agent A
"examples": [...], "evidence": {...}, ...
reuse conditions and dependencies
"guards" [{"type": "KeyUnique", ...}]
"deps" {"item": {"batch": 1, ...}, ...}
...
metric:Store revenue
another metric
feature:customer:spend_90d
a feature
...
more entries
check:KeyUnique(date_dim.d_date_sk)
one result, used by 3 definitions
"outcome" {"pass": true, "n_keys": 1096}
join:item|store_sales
revision 0 → 1
"on" [["ss_item_sk", "i_item_sk"]]
"filters" {"item": "i_is_current = 'Y'"}
"guards" [{"type": "KeyUnique"}], ...
check:KeyUnique(item.i_item_sk)
"filter" "i_is_current = 'Y'"
"outcome" {"pass": true, "n_keys": ...}
"deps": {"item": {"batch": 1}}, ...
3
Lookup
by name, versions
B valid
C item changed
N feature valid
pending
Maintain
6
rerun its checks
item key unique?
key 17 repeats
Repair
i_is_current='Y'
regression test
publish revision 1
else invalidate
4
Check and run
one snapshot
B SQL sketch
SUM(ss_net_paid)
WHERE i_category
= 'Electronics'
AND d_moy IN
(7, 8, 9) ...
checks pass
run SQL
N spend_90d
7
cutoff
90-day window
ETL job
loads item history
5
Database
table versions
change on writes
store_sales
batch 0
date_dim
batch 0
item
batch 0 → 1
i_item_sk i_category i_is_current
17 'Electronics' 'Y' current row
17 'Electronics' 'N' new history row
snapshot